

DBMS Model Question Paper

R23 | II Year - II Semester | 3 Hours | 70 Marks

Original Backlog Buddy practice resource - not an official university paper.

PART A - Answer all questions (10 x 2 = 20 marks)

1. Define data independence.
2. What is a weak entity?
3. State the purpose of a foreign key.
4. Define functional dependency.
5. Distinguish 3NF and BCNF.
6. State the ACID properties.
7. What is serializability?
8. Define a B+ tree index.
9. What is query optimization?
10. Define database recovery.

PART B - Answer one question from each unit (5 x 10 = 50 marks)

11. Explain three-schema architecture with a neat diagram. OR Design an ER model for a library.
12. Explain relational algebra operations. OR Write SQL queries using joins and subqueries.
13. Normalize a relation up to BCNF. OR Explain lossless and dependency-preserving decomposition.
14. Explain concurrency control using two-phase locking. OR Discuss recovery using logs.
15. Explain B+ tree operations. OR Compare static and dynamic hashing.

Use previous-paper patterns only for preparation; no question is guaranteed.